

FINAL SEMINAR • OCTOBER 2026

Multi-Page Visually Rich Document Understanding

A survey of system design and an empirical attribution of failure

Lewei Xu

Supervised by Dr. Yihao Ding • Dr. Daochang Liu



ONE DOCUMENT • 48 PAGES

THREE PAGES CARRY THE ANSWER

What this talk covers

20 minutes

Four parts

01 BACKGROUND

Why multi-page is a different problem

What a visually rich document is, and why the single-page setting does not simply scale up to many pages.

- Text, layout and pixels together
- Evidence sparse and scattered
- The system must choose what to read

02 LITERATURE

Organising the field

The review reads MP-VRDU as an evidence-management problem, sorted by where that evidence is handled.

- Four architectural families
- Text, visual, cross-page structure
- Every family relocates the bottleneck

03 EMPIRICAL STUDY

Locating the failure

The bulk of the talk. One pipeline, one stage moved at a time, so failure can be pinned to a stage.

- Representation, selection, reasoning
- Two benchmarks, one pipeline
- What fails, and what does not

04 CONCLUSION

Where this leads

A pipeline built straight out of the findings, and then the challenges that remain open.

- A simple adaptive-trajectory method
- What the thesis settles
- What it does not

TIME BUDGET

ONE TICK = ONE MINUTE

